

Logic Gates — Paper B

MEDIUM · NON-CALCULATOR · 35 MINUTES · 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. The mark for each question is in square brackets on the right.
- No calculator is needed anywhere on this paper.
- Truth-table rows go in **binary counting order**: 00, 01, 10, 11, and for three inputs 000, 001, 010, 011, 100, 101, 110, 111.
- Write expressions with AND, OR, NOT, NAND and NOR, and put **brackets wherever two different gates meet**. NOT applies only to the thing straight after it, so NOT A AND B means (NOT A) AND B.
- A reminder of the five gates is printed below. You may use it on every question.

REMINDER · THE FIVE GATES · YOU MAY USE THIS ON EVERY QUESTION

- AND** 1 only when **every** input is 1.
- OR** 1 when **at least one** input is 1. Both inputs being 1 still gives 1.
- NOT** One input. The output is the opposite of it.
- NAND** AND, then flip. 0 only when every input is 1; 1 on every other row.
- NOR** OR, then flip. 1 only when every input is 0; 0 on every other row.

SECTION 1 – GATES AND TABLES

1. Complete the truth table for a **three-input** AND gate and for a **three-input** OR gate. [2]

A	B	C	AND	OR
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

2. Answer these without drawing the tables.
- (a) A circuit has **4** inputs. How many rows does its truth table need? [1]
- (b) A circuit has **6** inputs. How many rows does its truth table need? [1]

(c) A truth table has **32** rows. How many inputs does the circuit have?

[1]

3. Each output column below belongs to one of AND, OR, NAND or NOR, and each gate is used once. Write the name of the gate in the bottom row of its column.

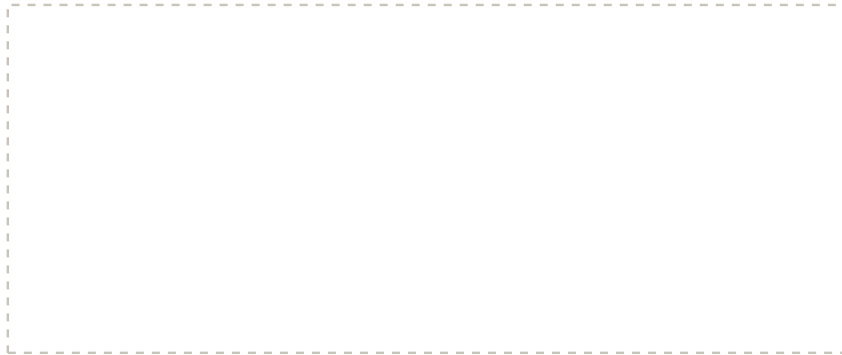
[4]

A	B	(i)	(ii)	(iii)	(iv)
0	0	1	0	1	0
0	1	1	0	0	1
1	0	1	0	0	1
1	1	0	1	0	1
Gate					

Count the 1s in each column, then find where the odd one out sits.

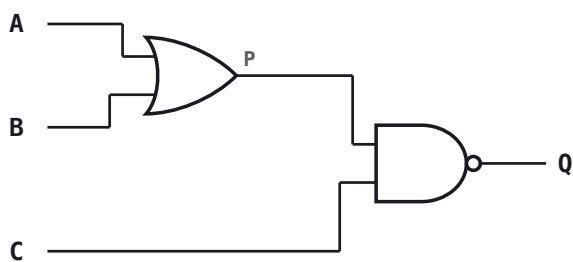
4. In the box, draw the standard symbol for (a) a NOT gate, (b) a NAND gate and (c) a NOR gate. Label each one, and show every input and output wire.

[3]



SECTION 2 – READING CIRCUITS

5. Study the circuit. The first gate gives the output P. Fill in the whole of column P before you start column Q.



- (a) Write the expression for Q.

[1]

(b) Complete the trace table.

[3]

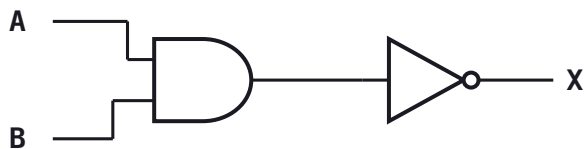
A	B	C	P	Q
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

(c) How many of the 8 rows have $Q = 1$?

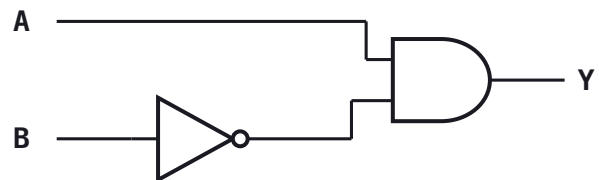
[1]

6. Look at these two small circuits.

(i)



(ii)



(a) Write the expression for X in circuit (i).

[1]

(b) Which single gate does circuit (i) behave as?

[1]

(c) Write the expression for Y in circuit (ii).

[1]

SECTION 3 – FROM WORDS TO CIRCUITS

7. A railway crossing near Tumakuru has a warning bell, Q. **A = 1** when a train is approaching. **B = 1** when the barrier is fully down. **C = 1** when the engineer presses the test switch. The bell rings when a train is approaching and the barrier is **not** fully down, **or** when the test switch is pressed.

(a) Write an expression for Q.

[2]

(b) Complete the truth table for Q.

[3]

A	B	C	Q
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

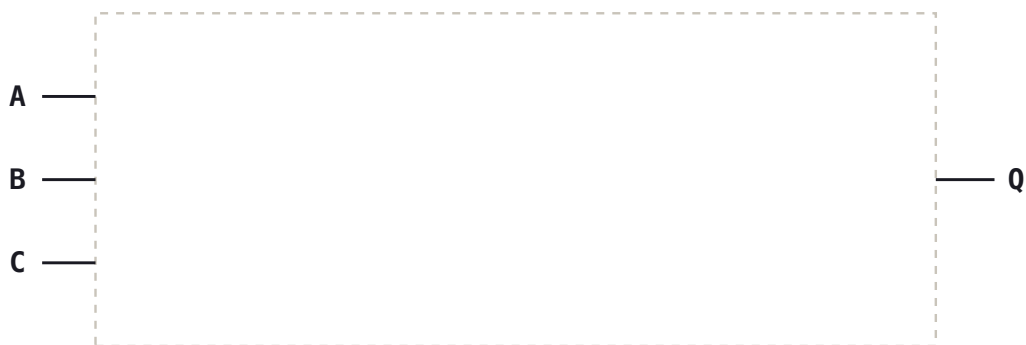
8. An ATM kiosk in Chennai has an alarm, Q. **A = 1** when the door is forced open. **B = 1** when the card slot is blocked. **C = 1** when the camera is covered. The alarm sounds when the door is forced, **or** when the card slot is blocked **and** the camera is covered.

(a) Write an expression for Q.

[2]

(b) Draw the circuit in the box, using the standard gate symbols.

[3]



End of paper. Check every truth table has all its rows in counting order, and that every expression has brackets wherever two different gates meet.